

Analysis of customer behavior and operational activities - Uber

Context:

Uber—a leading ride-hailing app—seeks to optimize operational efficiency and user experience management. Management has observed frequent fluctuations in trip cancellation rates and customer waiting times (VTAT/CTAT), alongside untapped revenue potential across different vehicle segments and time slots.

Note: AI-generated synthetic dataset created to simulate realistic Uber ride-sharing operations for 2024.

Core Objective:

Leverage the Uber trip operations dataset using Python, SQL, and Power BI to analyze operations, revenue, cancellation drivers, and customer segmentation, thereby formulating strategic recommendations to improve trip completion rates and optimize revenue.

1. Dataset Summary

Rows: 150000

Columns: 21

Key Features:

- Customer & Identification: Customer ID, Booking ID
- Time: Date, Time
- Trip & Location Details: Vehicle Type, Pickup Location, Drop Location, Ride Distance, Booking Status
- Financial & Payment: Booking Value, Payment Method
- Operational & Time Metrics: Avg VTAT, Avg CTAT
- Cancellation & Trip Failure: Cancelled Rides by Customer, Reason for cancelling by Customer, Cancelled Rides by Driver, Driver Cancellation Reason, Incomplete Rides, Incomplete Rides Reason.
- Feedback & Service Quality: Driver Ratings, Customer Rating.
- There are many duplicate Booking IDs and null values in multiple columns.

2. Exploratory Data Analysis using Python.

Overview:

- This dataset contains 150,000 records and 21 columns.
- Initial investigation reveals that 8 columns do not contain null values (Date, Time, BookingID, BookingStatus, Customer ID, Vehicle Type, Pickup Location, Drop Location). The remaining columns contain null values; however, this may be due to business logic. For example, the reason

for cancellation is only displayed when the trip is canceled by the customer or driver. Further investigation is needed on this issue.

- No two rows have the same value.

Details:

- After checking, I found no problem with the important columns having null values based on the Booking Status groups; it's just a logical business practice.

- After checking for duplicate data in the Booking ID column, I discovered 1233 duplicate entries. This data needs to be processed.

- I realized the need to add time-related fields to facilitate analysis. Therefore, I used the Date and Time columns to create new columns: Datetime, Year, Month, Day, Weekday, and Weekend.

- I created a data table in ascending order based on the Datetime column, and then handled duplicates by using the Booking ID of the most recent booking if two bookings were the same.

- After processing, only 148,767 records remain.

3. Data Analysis using SQL

Group 1: Operational Efficiency

- **Question 1:** Calculate the overall trip completion rate (Success Rate) and provide a detailed analysis by vehicle type. Which vehicle type has the highest rate of cancellations or "No Driver Found" instances?

	Total_Booking	Rate_Complete
1	92244	62.0057

- The trip completion rate for the entire dataset is 62%.

	'Vehicle Type'	Total_Booking	Total_Booking_2	Rate_Compeleted
1	Uber XL	2757	4414	62.4604
2	Bike	13927	22325	62.3830
3	Go Mini	18393	29549	62.2458
4	Premier Sedan	11153	17956	62.1129
5	eBike	6498	10477	62.0216
6	Auto	22961	37111	61.8711
7	Go Sedan	16555	26935	61.4628

- The completion rates for the different vehicle types are roughly similar, with Uber XL having the highest rate.

	'Vehicle Type' ▾	Total_Booking ▾	Total_Booking_2 ▾	Rate_Compeleted ▾
1	Go Sedan	6806	26935	25.2682
2	Bike	5588	22325	25.0302
3	Go Mini	7368	29549	24.9349
4	Premier Sedan	4476	17956	24.9276
5	Auto	9237	37111	24.8902
6	eBike	2607	10477	24.8831
7	Uber XL	1084	4414	24.5582

- Cancellation rates across vehicle types show little variation, hovering around 25%. The highest cancellation rate is for Go Sedan.

	'Vehicle Type' ▾	Total_Booking ▾	Total_Booking_2 ▾	Rate_No_Found ▾
1	Go Sedan	1943	26935	7.2137
2	Auto	2665	37111	7.1812
3	eBike	743	10477	7.0917
4	Uber XL	313	4414	7.0911
5	Premier Sedan	1272	17956	7.0840
6	Go Mini	1986	29549	6.7210
7	Bike	1492	22325	6.6831

- The rate of vehicle unavailability is similar across the board, ranging from 6% to 7%, with Go Sedan recording the highest figure at 7.2%.
- **Question 2:** How do the average vehicle allocation time (Avg VTAT) and average customer waiting time (Avg CTAT) vary across different times of the day (Morning, Midday, Afternoon, Evening)?

	Time_Of_Day ▾	'Booking Status' ▾	Avg_VTAT ▾	Avg_CTAT ▾
1	Early morning	Completed	8.45	29.84
2	Early morning	Cancelled by Driver	7.56	<null>
3	Early morning	No Driver Found	<null>	<null>
4	Early morning	Cancelled by Customer	12.55	<null>
5	Morning	Incomplete	6.01	20.03
6	Morning	Cancelled by Customer	12.6	<null>
7	Morning	Completed	8.53	30.06
8	Morning	Cancelled by Driver	7.52	<null>
9	Morning	No Driver Found	<null>	<null>

- The average time for drivers to reach the pickup location and the average waiting time remain relatively stable throughout the day; however, it is noticeable that trips cancelled by customers exhibit a higher Avg_VTAT compared to other statuses.
- **Question 3:** Identify the top 5 pickup routes with the highest volume of ride requests but the highest "No Driver Found" rates, in order to flag areas with a driver shortage.

	'Pickup Location'	Total_Booking
1	Old Gurgaon	83
2	Pataudi Chowk	79
3	Paharganj	78
4	Greater Noida	76
5	Vinobapuri	76

- Top 5 locations where customers place bookings but no drivers accept them—for example, 'Old Gurgaon', 'Pataudi Chowk', and 'Paharganj'.

Group 2: Revenue & Demand Analysis

- **Question 4:** What is the total revenue (Booking Value) generated from completed trips? Please provide a breakdown of the revenue contribution by vehicle type.

	'Vehicle Type'	Total_Amount
1	Auto	11631357
2	Go Mini	9330061
3	Go Sedan	8471253
4	Bike	7084312
5	Premier Sedan	5683485
6	eBike	3267699
7	Uber XL	1390891

- Revenue from completed trips amounted to 46,859,058 Rupees, with auto-rickshaws generating the highest revenue at 11,631,357 Rupees.
- **Question 5:** The relationship between ride distance and booking value. What is the average cost per kilometer (Booking Value / Ride Distance) for each vehicle type?

	'Vehicle Type'	Avg_Distance	Avg_Amount	Avg_Per_Trip
1	Bike	26	508.67	19.56
2	Go Mini	26.01	507.26	19.5
3	eBike	26.34	502.88	19.09
4	Auto	25.98	506.57	19.5
5	Premier Sedan	25.95	509.59	19.64
6	Uber XL	25.77	504.49	19.58
7	Go Sedan	25.98	511.7	19.7

- The average cost per trip is similar across vehicle types, indicating that there is no significant price difference between them.
- **Question 6:** Which payment method is most preferred for high-value trips (above average)?

	'Payment Method'	Total_Booking
1	UPI	17204
2	Cash	9528
3	Uber Wallet	4592
4	Credit Card	3803
5	Debit Card	3068

- UPI is the preferred payment method for high-value bookings.

Group 3: Root Cause & Experience Analysis

- **Question 7:** What are the primary reasons for trip cancellations by the customer and the driver?

	'Reason for cancelling by Customer'	Total_Booking
1	Wrong Address	2346
2	Change of plans	2323
3	Driver is not moving towards pickup location	2314
4	Driver asked to cancel	2276
5	AC is not working	1140

- Common reasons for customer-initiated cancellations include: wrong address, change of plans, the driver not moving toward the pickup location, or the driver asking to cancel.

	'Driver Cancellation Reason'	Total_Booking
1	Customer related issue	6774
2	The customer was coughing/sick	6692
3	Personal & Car related issues	6663
4	More than permitted people in there	6638

- Reasons for trip cancellations initiated by drivers typically include the following: customer-related issues, the customer coughing or appearing sick, personal or

vehicle-related issues, and the number of passengers exceeding the permitted limit.

- **Question 8: What is the primary reason for trips labeled "Incomplete" (Incomplete Rides Reason)?**

	`Incomplete Rides Reason`	Total_Booking
1	Customer Demand	3018
2	Vehicle Breakdown	2996
3	Other Issue	2929

- The trip was labeled "Incomplete" due to the following reasons: Customer Demand, Vehicle Breakdown, or Other Issue.

- **Question 9: Determine the correlation between driver ratings and customer ratings. Do trips with long average wait times (Avg VTAT) tend to receive low ratings?**

	Corr_Driver_Customer	Corr_VTAT_Customer
1	-0.0004407807696989555	-0.004320815129778595

- The Pearson correlation coefficient between Avg VTAT and Customer Rating is -0.004, indicating no linear relationship. The time spent waiting for vehicle dispatch does not affect the customer's final satisfaction level on the app.

Group 4: Time-Series & Loyalty Analysis

- **Question 10:** Identify the peak hours and day of the week with the highest ride-booking demand.

	Time_Group	Total_Booking
1	18	12288
2	17	10964
3	19	10961
4	16	9562
5	20	9547
6	10	9498
7	11	8322
8	9	8163
9	15	8131

- The 5:00 PM to 7:00 PM time slot is the peak period.

	Time_Group		Total_Booking
1		18	881
2		19	758
3		17	743
4		16	692
5		20	667
6		10	644
7		11	595
8		9	585
9		21	549

- This time slot also frequently sees bookings where no driver is found.

	Weekday		Total_Booking
1		0	21466
2		5	21381
3		6	21234
4		2	21223
5		4	21213
6		1	21210
7		3	21040

- The demand for booking rides is the same across weekdays.
- **Question 11:** Segment customers based on booking frequency and total spending. Identify Customer IDs exhibiting "mass cancellation" behavior (canceling more than two consecutive trips) for inclusion in the system's alert list.

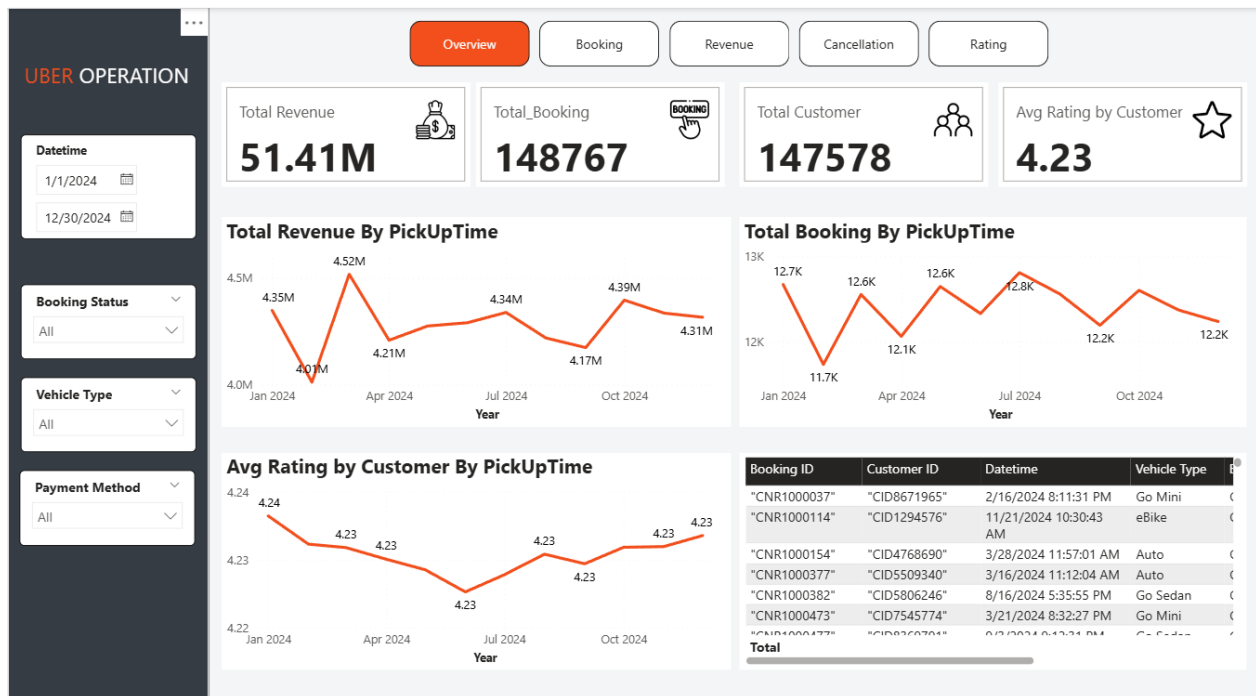
	Customer_Segment1		Total_Customer
1	Loyal Customer		5
2	Regular		869
3	New / Low Value		146704

- This is the number of customers in each segment.

Customer ID

- No customer has cancelled more than two consecutive trips.

4. Dashboard in Power BI



5. Business Recommendations

1. Solutions to Improve Completion Rates

A. Driver Matching & Supply Management

- **Driver Relocation & Supply Dispatch at Hotspots:**
 - Implement a **Dynamic Incentive** policy (per-trip bonuses) for drivers navigating to heavy shortage areas such as *Old Gurgaon, Pataudi Chowk, Paharganj, Greater Noida, and Vinobapuri*.
 - Trigger real-time **Heatmap Alerts** on the driver app during the peak window (**17:00 – 19:00**) to concentrate supply.
- **Matching Algorithm Optimization:**
 - Data reveals that average vehicle allocation time (Avg_VTAT) for customer-cancelled rides is noticeably higher (~12.5 minutes compared to ~8.5 minutes for completed rides). The initial driver dispatch radius should be capped to reduce VTAT, preventing long-distance matches that lead to driver non-movement or requests for customer cancellations.

B. Mitigation of Customer & Driver Cancellations

- **Address Pickup Location Issues ("Wrong Address"):**
 - Integrate an **Auto-Suggest Pickup Snap** feature and mandate visual confirmation from the customer before placing a ride request to mitigate this top cancellation cause.
- **Penalize Inactive Drivers / Forced Cancellations:**
 - Implement automatic penalties and credit score reductions for drivers who accept rides but remain stationary ("Driver not moving") or prompt customers to cancel ("Driver asked to cancel").
- **Transport Policies & Quality Standards:**
 - Add a passenger count selector in the app to prevent driver rejections due to exceeding capacity limits.
 - Establish regular AC inspection protocols for vehicles (especially Go Sedan and Premier) as "AC not working" is a frequent reason for cancellation.

2. Solutions to Optimize Revenue

A. Segment-Based Price Restructuring (Dynamic Pricing Model)

- **Price Differentiation Across Vehicle Tiers:**
 - Currently, the average rate per kilometer exhibits negligible variation between Bike (19.56) and Premier Sedan / Uber XL (19.64 – 19.7). This underutilizes the potential yield of premium and larger vehicle segments.
 - **Proposal:** Increase the base fare and per-kilometer rate for Premier Sedan and Uber XL while maintaining competitive entry pricing for Auto and Bike. This will raise take-rates per trip in premium tiers without dampening overall demand.
- **Smart Surge Pricing:**
 - Apply marginal dynamic fare increases during the peak hours of **17:00 – 19:00**. The revenue surplus directly boosts top-line earnings while incentivizing drivers to stay online, thereby suppressing "No Driver Found" rates.

B. Payment Optimization & High-Value Customer Activation

- **Promote UPI Integration & Co-Promotions:**
 - Insights show that **UPI** is the predominant payment method for high-value transactions.
 - Partner with UPI platforms and banks to offer cashback and discount campaigns on long-distance trips or Sedan/XL bookings to encourage higher basket sizes.
- **Loyalty & Retention Programs:**

- The "Loyal Customer" segment currently makes up a negligible fraction (5 customers), while "New / Low Value" constitutes the vast majority.
- Build a **UberRewards / Membership Tier** scheme with redeemable points and vouchers to convert low-frequency users into loyal customers, increasing Customer Lifetime Value (LTV).